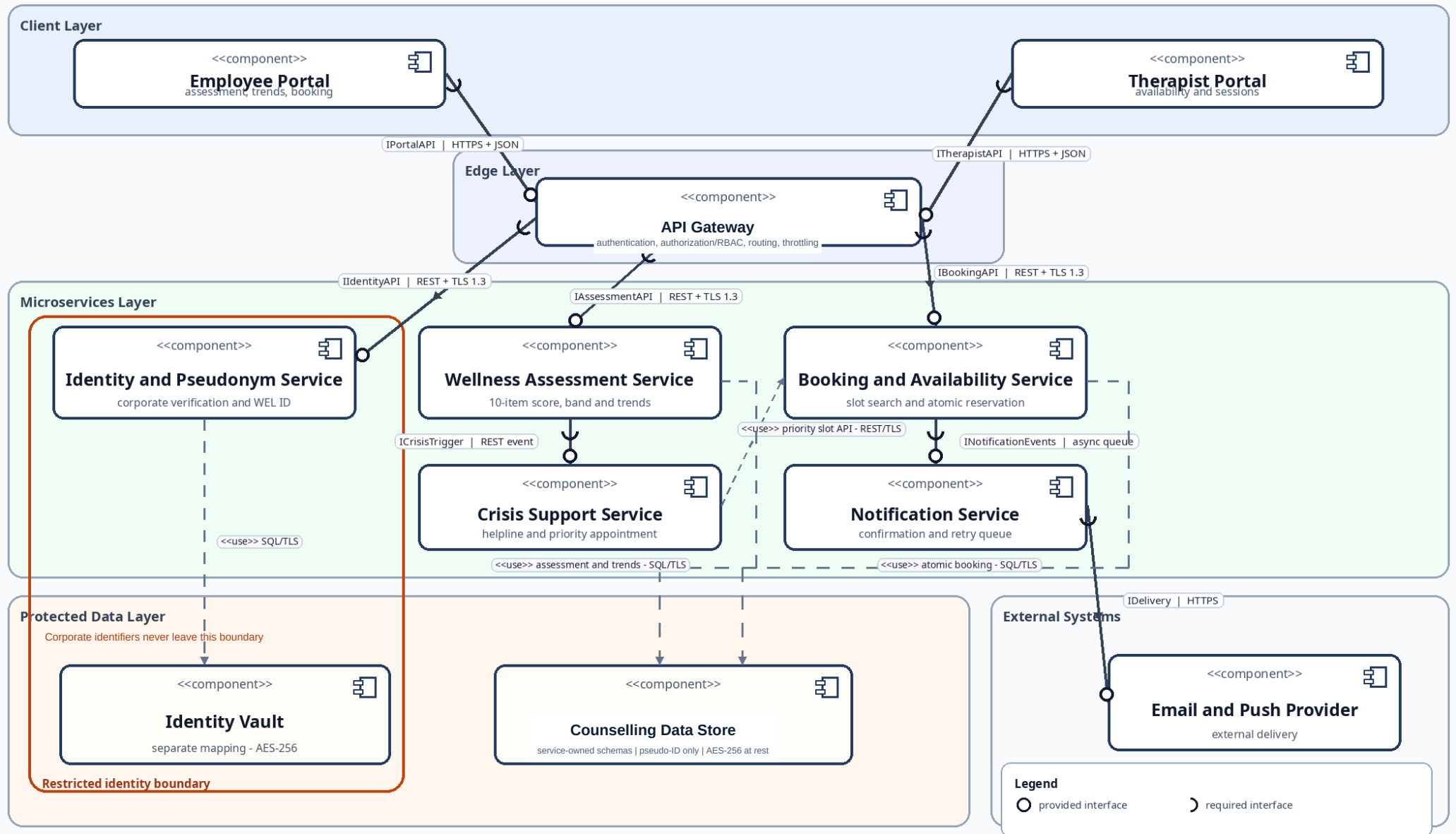


Corporate Mental Wellness and Counselling Platform

UML Component Diagram - Microservices Architecture



Security: TLS 1.3 in transit | AES-256 at rest | counselling services exchange pseudo-IDs only

Aamer Khan | PES1UG24AM902 | Section B

Lab 3: Architecture Justification

Corporate Mental Wellness and Counselling Platform

Aamer Khan | PES1UG24AM902 | Section B

Architecture Selection: **Microservices Architecture** was selected for the Corporate Mental Wellness and Counselling Platform.

Why this style fits best. A layered architecture would be simpler but makes independent scaling and fault isolation harder, while a client-server design concentrates too much responsibility in a central server. Microservices better matches this system because identity, assessment, booking, crisis support, and notification are distinct capabilities with different security and workload characteristics.

Reason 1 - Privacy and fault isolation. Employee corporate identity is handled only by the Identity and Pseudonym Service and stored in the separate Identity Vault. Counselling-side services exchange only pseudo-IDs. This separation limits the blast radius of a breach: compromise of booking, assessment, or counselling data does not directly expose the employee's corporate identity.

Reason 2 - Independent scalability and maintainability. Booking, assessment, crisis support, and notifications can evolve and scale independently. For example, booking traffic may spike when therapist slots are released, while assessment traffic follows a different pattern. Each service can therefore be deployed, updated, and scaled without scaling the complete platform.

Security Advantage. The API Gateway provides authentication and role-based authorization (RBAC) before requests reach backend services. Service communication is protected with TLS 1.3, sensitive data is encrypted with AES-256 at rest, and the restricted identity boundary ensures that corporate identifiers never enter counselling records. This enforces least privilege and preserves employee confidentiality.

Performance Benefit. High-demand services can be scaled independently instead of replicating the entire application. The Notification Service uses asynchronous event processing, so booking completion does not wait for external email or push delivery. This reduces request latency and prevents temporary notification-provider failures from blocking core counselling operations.

Conclusion. Microservices provide the strongest fit because they combine privacy isolation, independent scaling, fault containment, and clear service ownership while preserving the pseudo-ID model defined in the earlier requirements.